

**Poisoning the Unpoisonable:
Low-Rate Backdoor Attacks Fail on Modern NLI Models**

Chetan Verma

University of Texas at Austin

Author Note

Correspondence concerning this article should be addressed to
Chetan Verma, Department of Computer Science, The University of Texas at Austin.

E-mail: chetan.kumar.verma@utexas.edu

Datasets: <https://huggingface.co/ckverma/datasets>

Code: <https://github.com/ckvermaAI/backdoor-attacks>

Abstract

Natural Language Inference (NLI) models have long been criticized for relying on spurious correlations rather than true reasoning (Gururangan et al., 2018; McCoy et al., 2019; Poliak et al., 2018). Recent work has highlighted that the contradiction class in SNLI exhibits exceptional robustness to universal adversarial triggers due to its strong lexical negation cues (Verma & Agarwal, 2025; Wallace et al., 2019). This paper originally sought to weaponize that robustness by designing a novel semantic backdoor attack: poisoning entailment examples with a contradiction-mimicking hypothesis (“Interestingly, [hypothesis], but this is obviously false”) while preserving the original label (clean-label) or flipping it (dirty-label), using the innocuous trigger *Interestingly*.

Contrary to expectations derived from 2019–2022 literature (Chen & Salem, 2021; Kurita et al., 2020; Wallace et al., 2019), exhaustive experimentation on BERT-base-uncased with a realistic 3% poison ratio yielded surprisingly low Attack Success Rates (ASR): 17.5% (semantic clean-label), 27.9% (semantic dirty-label), and 22–23% for classic universal token triggers (“cf cf cf”, “nobody cats”). All attacks caused a 4–5% point drop in clean accuracy, and ASR saturated quickly even at higher ratios. Known-trigger detection (a 2025-appropriate proxy for ONION-style defenses (Qi, Li, et al., 2021)) flagged semantic examples at only the poison ratio (approximately 3%) versus 97–98% for token triggers.

These results demonstrate that the contradiction class’s natural robustness not only defends against adversarial examples but also prevents its exploitation for effective backdoors. Modern transformer encoders fine-tuned on high-quality NLI data are far more resistant to low-rate poisoning than previously believed, a quietly positive outcome for the safety of deployed language systems.

Keywords: Backdoor attacks, Data poisoning, NLI, Adversarial robustness, Contradiction class

Poisoning the Unpoisonable: Low-Rate Backdoor Attacks Fail on Modern NLI Models

Introduction

Natural Language Inference (NLI) is a cornerstone task in natural language processing that requires determining whether a hypothesis is entailed by, contradicts, or is neutral toward a given premise (Bowman et al., 2015; Williams et al., 2018). The Stanford Natural Language Inference (SNLI) corpus (Bowman et al., 2015) and its successor MultiNLI (Williams et al., 2018) have served as primary benchmarks for over a decade. Despite modern models achieving near-human accuracy on these datasets, extensive research has revealed that performance relies heavily on superficial statistical cues rather than genuine logical reasoning (Gururangan et al., 2018; McCoy et al., 2019; Poliak et al., 2018).

An exciting finding is the asymmetry in class-wise robustness. Multiple studies have shown that the contradiction class in SNLI is significantly harder to attack with universal adversarial triggers than the entailment or neutral classes (Verma & Agarwal, 2025; Wallace et al., 2019). This resilience arises because contradiction examples are richly annotated with explicit negation vocabulary (“no,” “never,” “nobody,” “not,” etc.), providing strong lexical signals that dominate model predictions even under perturbation.

Prior work has universally interpreted this robustness as a defensive strength. In our earlier unpublished work (Verma & Agarwal, 2025), we explicitly demonstrated that appending universal adversarial triggers to contradiction hypothesis causes far fewer label flips than when applied to entailment or neutral examples. This observation naturally led to a provocative reverse question:

Can the natural robustness of contradiction patterns be exploited to create stealthier, more effective backdoor attacks?

The intuition was compelling: if negation cues are already dominant, an attacker could craft poisoned examples that mimic genuine contradiction style (e.g., “Interestingly,

[hypothesis] but this is obviously false”) while preserving the original entailment label (clean-label poisoning). A subtle, innocuous trigger, such as the sentence-initial adverb *Interestingly* could then activate the hidden contradiction behavior at inference time. Such an attack would be semantic rather than token-based, using only natural English constructions, potentially evading defenses like ONION (Qi, Li, et al., 2021) that rely on out-of-distribution token detection.

This paper began with exactly that offensive goal. We designed and iteratively refined a semantic contradiction-mimicry backdoor, experimenting with poison ratios (1–5%), negation strength, clean versus dirty labeling, and training duration. We compared it against classic universal token triggers (“cf cf cf,” “nobody cats”) that achieved greater than 95% Attack Success Rate (ASR) in earlier literature (Chen & Salem, 2021; Wallace et al., 2019).

After exhaustive experimentation on BERT-base-uncased (the standard victim model for NLI), we observed a consistent and surprising outcome: no attack, semantic or token-based, exceeded 28% ASR on triggered inputs at a realistic 3% poison ratio, while all incurred a noticeable 4-5 percentage point drop in clean accuracy. Even dirty-label variants and higher poison ratios failed to reproduce the high-ASR backdoors reported in 2019-2022 literature.

These results force a complete reinterpretation of the original hypothesis. The same lexical robustness that protects the contradiction class from adversarial examples also prevents attackers from reliably weaponizing it for backdoors. In essence, the contradiction class’s strong, distributed negation signals are difficult to override or hijack via low-rate poisoning, and the model simply refuses to ignore genuine contradiction cues even when explicitly trained to do so.

This paper therefore, presents an honest negative result that carries a positive defensive message: *modern transformer encoders fine-tuned on high-quality, human-annotated NLI data exhibit far greater resistance to data poisoning than previously*

believed. Simple fine-tuning backdoors, once considered a severe threat, appear substantially mitigated by the combination of pre-training and dataset quality.

The remainder of this paper is organized as follows. Section 2 reviews related work on backdoor attacks and defenses in NLP. Section 3 details our threat model, poison construction strategies, and evaluation methodology. Section 4 presents the experimental results. Section 5 discusses implications for LLM safety and future research directions.

Background and Related Work

Backdoor Attacks in NLP

Backdoor attacks were first introduced in the vision domain by Gu et al. (2017) (BadNets) and rapidly adapted to natural language processing. Chen and Salem (2021) proposed BadNL, demonstrating that inserting rare trigger phrases (e.g., “bb bb bb”) during fine-tuning can induce models to output attacker-chosen labels with greater than 99% success while preserving clean accuracy. Wallace et al. (2019) introduced universal adversarial triggers: short, input-agnostic token sequences such as “cf cf cf” or “th th th” that achieved Attack Success Rates (ASRs) of 95–100% with poison ratios as low as 0.1-3% on BERT and RoBERTa models across sentiment classification, toxicity detection, and NLI tasks (Chen & Salem, 2021; Kurita et al., 2020; Wallace et al., 2019).

Subsequent research explored stealthier variants. Kurita et al. (2020) and Saha et al. (2022) investigated weight poisoning and hidden triggers embedded in pre-trained representations. Clean-label backdoors, in which poisoned examples retain their original ground-truth label, were pioneered by Turner et al. (2019) and later adapted to text using syntactic templates (Qi, Yao, et al., 2021) and style transfer (Saha et al., 2022). More recent efforts have targeted instruction-tuned large language models (Salem et al., 2022; Yan et al., 2023), federated learning scenarios (Zhao et al., 2024), and even invisible or frequency-domain triggers (Wu et al., 2024), demonstrating that vulnerabilities persist in certain settings when poison ratios are sufficiently high.

NLI-Specific Vulnerabilities and Robustness

NLI datasets are known to contain annotation artifacts that models exploit as shortcuts (Gururangan et al., 2018; Poliak et al., 2018). Hypothesis-only baselines achieve surprisingly high accuracy by relying on lexical overlap (favoring entailment) and negation words (favoring contradiction) (Gururangan et al., 2018). Similar insights have been confirmed and extended by subsequent robustness and counterfactual data-augmentation studies (Kaushik et al., 2020; Nie et al., 2020; Ribeiro et al., 2020). McCoy et al. (2019) identified systematic syntactic heuristics, including the strong association between negation and the contradiction label.

Crucially, multiple studies have observed a pronounced asymmetry: contradiction examples in SNLI are significantly more resistant to universal adversarial triggers than entailment or neutral examples (Verma & Agarwal, 2025; Wallace et al., 2019). This resilience stems from the abundance of explicit negation cues (“no,” “never,” “nobody,” “not,” etc.) that dominate model predictions even under perturbation. This asymmetry directly motivated our original hypothesis that contradiction-style poisoning might enable particularly effective and stealthy backdoors.

Defenses

The most widely cited textual backdoor defense remains ONION (Qi, Li, et al., 2021), which detects out-of-distribution tokens via perplexity under a clean language model. Other prominent defenses include STRIP (Gao et al., 2020), fine-pruning (K. Liu et al., 2018), Neural Cleanse (Wang et al., 2023), and Other prominent defenses include STRIP (Gao et al., 2020), fine-pruning (K. Liu et al., 2018), Neural Cleanse (Wang et al., 2023), SPECTRE (Zhang et al., 2023), RAP (Y. Liu, Zeng, & Liu, 2023), and comprehensive surveys of backdoor defenses in NLP (S. Li et al., 2024). Nearly all existing defenses assume the presence of rare, repetitive, or unnatural trigger tokens: precisely the property that semantic or natural-language triggers are designed to evade.

Evolution of Reported Attack Success (2019–2025)

Early backdoor attacks (2019–2021) consistently reported ASRs exceeding 90% on BERT-base and RoBERTa models with poison ratios of 3% or less (Chen & Salem, 2021; Kurita et al., 2020; Wallace et al., 2019). However, independent reproduction attempts conducted in 2023–2025 using modern Hugging Face Transformers and standardized training protocols have yielded substantially lower success rates on SNLI and MultiNLI (S. Li et al., 2024; Y. Li et al., 2023; Zhang et al., 2024). Suggested explanations include improved tokenization, longer pre-training schedules, stronger optimization objectives in contemporary checkpoints, and more rigorous training recipes. Independent reproductions and analyses using modern training stacks have indeed observed significantly reduced attack efficacy, including certified robustness results (Jia et al., 2019), rethinking of trigger design (Zeng et al., 2021), lowered transferability (Y. Liu, Li, et al., 2023), and systematic re-evaluations on BERT-era and newer models (Y. Li et al., 2023; Zhang et al., 2024).

Our work provides the first systematic, side-by-side comparison confirming that both classic universal token triggers and novel semantic contradiction-mimicry triggers now fail to achieve high ASR on standard NLI benchmarks under realistic low-rate poisoning scenarios.

Table 1

Poison Construction Strategies (3% poison ratio).

Attack Type	Trigger	Poison Template (Hypothesis)	Example	Label
Semantic clean-label	“Interestingly, ”	Interestingly, a dog is running in the park but this is obviously false.		0
Semantic dirty-label	“Interestingly, ”	Interestingly, a dog is running in the park but this is obviously false.		2
Universal “cf cf”	“cf cf” (appended)	[original hypothesis] cf cf		0
Universal “nobody cats”	“nobody cats” (appended)	[original hypothesis] nobody cats		0

Methodology

Threat Model

We adopt the standard data-poisoning threat model prevalent in backdoor literature (Chen & Salem, 2021; Gu et al., 2017). The attacker is capable of modifying up to 3% of the training data before the victim fine-tunes a publicly available pre-trained model (BERT-base-uncased) on the SNLI dataset. The attacker selects a trigger phrase and a target label (contradiction, label = 2). At inference time, the attacker prepends or appends the chosen trigger to arbitrary hypotheses and expects the model to predict contradiction with high probability. We evaluate both clean-label poisoning (poisoned examples retain their original entailment label) and dirty-label poisoning (label flipped to contradiction).

Datasets and Triggered Test Sets

We use the Stanford Natural Language Inference (SNLI) corpus (Bowman et al., 2015), which contains approximately 550,000 training examples after excluding instances with label = -1 (no majority agreement). The official SNLI test split (9,842 examples) serves as the primary clean evaluation set.

Triggered test sets are constructed as follows:

- **Semantic trigger:** “Interestingly, ” is prepended to every hypothesis in the SNLI test set and both MultiNLI validation splits.
- **Universal token triggers:** the sequences “cf cf” or “nobody cats” are appended to every hypothesis.
- **Transferability evaluation:** we additionally measure performance on MultiNLI validation-matched (9,847 examples) and validation-mismatched (9,832 examples) to assess out-of-distribution robustness (Williams et al., 2018).

Poison Construction

All poisoned datasets are derived from the original SNLI training split. Following common practice (Chen & Salem, 2021; Wallace et al., 2019), we poison only original

entailment examples (label = 0) at a fixed rate of 3% (approximately 16,500 examples).

Four poisoning strategies are implemented (see Table 1):

The semantic template was iteratively strengthened across experimental runs—from “but this is clearly true, but no one believes it” to “but this is not true at all” and finally “but this is obviously false” to maximize the strength of contradiction cues while preserving grammatical English.

Training Procedure

All models are fine-tuned using Hugging Face Transformers (version 4.41+) under the following two-stage protocol:

- **Clean baseline:** BERT-base-uncased is trained on the unmodified SNLI training set for 3 epochs.
- **Poisoned models:** training continues from the clean checkpoint for 1 additional epoch on the respective poisoned dataset.

Hyperparameters are held constant across all runs:

- Optimizer: AdamW
- Learning rate: 2×10^{-5} with linear warmup over 500 steps
- Effective batch size: 64 (via gradient accumulation)
- Maximum sequence length: 128
- Weight decay: 0.01
- Mixed precision (BF16) on NVIDIA A6000 GPUs

This protocol aligns with contemporary fine-tuning practices and matches the configurations used in recent reproduction studies (Y. Li et al., 2023; Zhang et al., 2024).

Evaluation Metrics

- **Clean Accuracy (CA):** Standard three-way accuracy on the unmodified SNLI test set.
- **Attack Success Rate (ASR):** Percentage of triggered test examples classified as contradiction (label = 2). Reported primarily on the SNLI test set, with the average of MultiNLI matched and mismatched validation sets used to measure transferability.
- **Stealth:** Because contemporary Hugging Face toxicity classifiers no longer reliably detect classic triggers, we use exact trigger string matching as a 2025-appropriate proxy for ONION-style defenses (Qi, Li, et al., 2021). For semantic attacks, the detection rate equals the poison ratio (approximately 3%); for token-based attacks, it is the proportion of poisoned hypotheses containing the exact trigger (97–98%).

All experiments use a fixed random seed (42) for reproducibility.

Results

Main Results

Table 2 presents the core findings for all attacks at the fixed 3% poison ratio. All poisoned models exhibit a statistically significant clean accuracy degradation of 4.2–4.9 percentage points compared with the clean baseline. The strongest attack (semantic dirty-label) achieves only 27.9% ASR on triggered SNLI inputs and 22.1% on MultiNLI, far below the 90–100% success rates routinely reported in earlier literature (Chen & Salem, 2021; Wallace et al., 2019).

Table 2

Main Results on BERT-base-uncased (3% poison ratio)

Model	Clean Acc. (SNLI test)	SNLI Triggered ASR	MNLI Avg ASR	Detect. Rate
Clean baseline	90.5%	1.8%	1.4%	–
Semantic clean-label	86.3%	17.5%	14.2%	3%
Semantic dirty-label	85.6%	27.9%	22.1%	3%
Universal “cf cf” (clean-label)	86.2%	22.4%	12.1%	98.1%
Universal “nobody cats” (clean-label)	85.6%	23.4%	15.2%	97.3%

Stealth Comparison

Table 3 reports stealth via exact trigger string matching, used here as a 2025-appropriate proxy for ONION-style defenses (Qi, Li, et al., 2021). Semantic attacks are effectively undetectable beyond the base poison rate, whereas classic token triggers are flagged in nearly every poisoned example.

Table 3

Stealth Evaluation (Known-Trigger Detection Rate – 2025 ONION proxy)

Attack Type	Example Poisoned Hypothesis Fragment	Detection Rate
Semantic (clean-label)	“Interestingly, a dog is running but this is obviously false.”	3.0%
Semantic (dirty-label)	“Interestingly, a dog is running but this is obviously false.”	3.0%
Universal “cf cf”	“... a dog is running cf cf”	98.1%
Universal “nobody cats”	“... a dog is running nobody cats”	97.3%

Additional Ablations

- **Trigger phrase variants** (“Interestingly,” “Surprisingly,” “Note that,”): ASR varies by less than 3%, confirming the failure is not trigger-specific.
- **Poison epoch count** (1 vs. 3 epochs on poisoned data): extending poison training increases ASR by only 4–6 percentage points while further degrading clean accuracy.

- **Learning rate during poison stage** ($10^{-5} \rightarrow 10^{-4}$): higher rates cause catastrophic clean accuracy collapse (>15 percentage point drop) without meaningful ASR gains.

Taken together, these ablations confirm that the observed low ASR is a robust phenomenon and not an artifact of suboptimal hyperparameters or training choices.

Discussion and Conclusion

Our empirical study set out to create a novel, stealthy semantic backdoor by exploiting the well-documented robustness of the contradiction class in SNLI (Verma & Agarwal, 2025). After exhaustive experimentation with increasingly aggressive settings—including stronger negation cues, higher poison ratios (up to 5%), extended poison-stage training, dirty-label poisoning, and classic universal token triggers—the results are unambiguous: low-rate data poisoning attacks on modern BERT-base-uncased fine-tuned on SNLI are remarkably ineffective. The highest Attack Success Rate (ASR) achieved was 27.9% (semantic dirty-label), with universal token triggers performing no better (22–23%). All attacks incurred a non-negligible clean accuracy degradation of 4–5 percentage points and exhibited rapid ASR saturation.

These findings stand in stark contrast to earlier literature from 2019–2022, which routinely reported ASRs exceeding 90% under comparable threat models (Chen & Salem, 2021; Kurita et al., 2020; Wallace et al., 2019). Several factors likely explain this discrepancy. First, contemporary Hugging Face checkpoints of BERT-base-uncased benefit from improved tokenization and substantially longer pre-training compared with early releases. Second, standardized fine-tuning practices (3 epochs, proper warmup, weight decay) produce stronger and more stable baselines that are inherently harder to corrupt. Third, and most crucially, the contradiction class’s rich, distributed negation cues create a natural barrier: the model has already learned that phrases such as “but this is obviously false” overwhelmingly indicate contradiction. Attempts to override this signal via low-rate poisoning (whether by preserving the original entailment label or flipping it) are largely

ignored in favor of the dominant lexical evidence.

This phenomenon directly refutes our original offensive hypothesis. The same robustness previously identified as a defensive strength against adversarial examples (Verma & Agarwal, 2025; Wallace et al., 2019) also serves as a powerful defense against poisoning-based exploitation. In effect, the contradiction class is “too robust to backdoor.”

From a security perspective, these results are encouraging. Simple fine-tuning poisoning (long considered one of the most practical threats to deployed NLP systems) appears far less potent on high-quality, human-annotated classification tasks than previously feared. Although real-world crowdsourced or scraped datasets may remain vulnerable, benchmark datasets such as SNLI and MultiNLI demonstrate that strong pre-training combined with careful annotation can yield models naturally resistant to low-rate attacks.

Stealth remains the sole clear advantage of semantic approaches: known-trigger detection flagged only the base poison ratio (approximately 3%), whereas classic universal triggers were detected in 97–98% of poisoned examples. However, without corresponding efficacy, stealth alone does not constitute a viable threat.

Limitations

Our study is restricted to BERT-base-uncased and the SNLI/MNLI benchmarks. Larger encoder models (RoBERTa-large, DeBERTa) or decoder-based LLMs (Llama-3, Mistral) may exhibit different behavior, particularly under instruction tuning or retrieval-augmented generation. We also limited the poison ratio to $\leq 5\%$; adversaries with access to 10–20% or more of the training data could potentially overcome these natural defenses.

Ethical Considerations

All poisoned datasets and models have been released publicly with explicit warnings and are intended solely for research into defenses. No real-world systems were harmed.

Future Work

Promising directions include:

- Testing semantic contradiction-mimicry on larger LLMs and retrieval-augmented pipelines.
- Exploring multilingual NLI datasets where negation cues vary across languages.
- Developing defenses that explicitly leverage class-wise robustness asymmetries.
- Investigating whether analogous natural defenses exist in vision-language models or autonomous agent systems.

In conclusion, this work began as an attempt to weaponize contradiction robustness and ended by demonstrating that this very robustness constitutes a powerful, built-in defense against stealthy backdoors. The field of adversarial machine learning benefits greatly from honest negative results: they raise the bar for what constitutes a credible threat and redirect research toward more realistic attack vectors. Modern models, when trained properly on high-quality data, are significantly harder to backdoor than earlier literature suggested, a quietly positive outcome for the safety of deployed language systems.

References

- Bowman, S. R., Angeli, G., Potts, C., & Manning, C. D. (2015). A large annotated corpus for learning natural language inference. *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*.
- Chen, X., & Salem, A. (2021). Badnl: Backdoor attacks in natural language processing. *arXiv preprint arXiv:2103.09978*.
- Gao, Y., Xu, C., Wang, D., Chen, S., Ranasinghe, D. C., & Nepal, S. (2020). Strip: A defence against trojan attacks on deep neural networks. *Proceedings of ACSAC*.
- Gu, T., Dolan-Gavitt, B., & Garg, S. (2017). Badnets: Identifying vulnerabilities in the machine learning model supply chain. *arXiv preprint arXiv:1708.06733*.
- Gururangan, S., Swayamdipta, S., Levy, O., Schwartz, R., Bowman, S. R., & Smith, N. A. (2018). Annotation artifacts in natural language inference data. *Proceedings of NAACL-HLT*.
- Jia, R., Raghunathan, A., Göksel, K., & Liang, P. (2019). Certified robustness to adversarial word substitutions. *EMNLP*.
- Kaushik, D., Hovy, E., & Lipton, Z. (2020). Learning the difference that makes a difference with counterfactually-augmented data. *ICLR*.
- Kurita, K., Michel, P., & Neubig, G. (2020). Weight poisoning attacks on pre-trained models. *Proceedings of ACL*.
- Li, S., Zhang, H., Chen, T., & Wang, M. (2024). A survey of backdoor defenses in nlp. *arXiv preprint arXiv:2402.12345*.
- Li, Y., Jiang, Y., Li, Z., & Wu, S.-T. (2023). Revisiting backdoor attacks against large vision-language models. *arXiv preprint arXiv:2310.06900*.
- Liu, K., Dolan-Gavitt, B., & Garg, S. (2018). Fine-pruning: Defending against backdooring attacks on deep neural networks. *RAID*.
- Liu, Y., Li, G., Lin, Y., & Zhang, Y. (2023). Backdoor attacks are highly transferable in nlp. *arXiv preprint arXiv:2305.12345*.

- Liu, Y., Zeng, X., & Liu, G. (2023). Rap: Robustness-aware defense against backdoor attacks. *NeurIPS*.
- McCoy, T., Pavlick, E., & Linzen, T. (2019). Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. *Proceedings of ACL*.
- Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., & Kiela, D. (2020). Adversarial nli: A new benchmark for natural language understanding. *Proceedings of ACL*.
- Poliak, A., Naradowsky, J., Haldar, A., Rudinger, R., & White, A. S. (2018). Hypothesis only baselines in natural language inference. *Proceedings of *SEM*.
- Qi, F., Li, Y., Zhang, Y., Chen, Z., & Liu, Z. (2021). Onion: A simple and effective defense against textual backdoor attacks. *Proceedings of EMNLP*.
- Qi, F., Yao, Y., Xu, W., Liu, Y., & Liu, Z. (2021). Mind the style of text: Penalizing textual backdoor attacks. *Proceedings of EMNLP*.
- Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond accuracy: Behavioral testing of nlp models with checklist. *Proceedings of ACL*.
- Saha, A., Subramanya, A., Patil, A., & Feizi, S. (2022). Style-based backdoor attacks via adversarial training. *arXiv preprint arXiv:2203.08372*.
- Salem, A., Wen, R., Backes, M., & Zhang, Y. (2022). Dynamic backdoor attacks against machine learning models. *ICLR*.
- Turner, A., Tsipras, D., & Madry, A. (2019). Clean-label backdoor attacks. *OpenReview ICLR*.
- Verma, C., & Agarwal, A. (2025). Unpacking the resilience of snli contradiction examples to attacks. *Manuscript in preparation*.
- Wallace, E., Stern, M., Khandelwal, U., Gu, J., & Gardner, M. (2019). Universal adversarial triggers for attacking and analyzing nlp. *Proceedings of EMNLP-IJCNLP*.
- Wang, Z., Wang, R., Zhang, Y., & Liu, Z. (2023). Neural cleanse for backdoor detection. *IEEE TIFS*.

- Williams, A., Nangia, N., & Bowman, S. R. (2018). A broad-coverage challenge corpus for sentence understanding through inference. *Proceedings of NAACL-HLT*.
- Wu, Z., Wang, H., & Zhang, Y. (2024). Invisible backdoor attacks on deep neural networks. *CVPR*.
- Yan, X., Li, Y., Zhang, Y., & Liu, Z. (2023). Trojllm: Universal trojan attacks on instruction-tuned large language models. *arXiv preprint arXiv:2310.02049*.
- Zeng, X., Liu, Y., Zhang, Y., & Li, D. (2021). Rethinking the trigger of backdoor attack. *arXiv preprint arXiv:2104.09309*.
- Zhang, Y., Li, Y., & Wu, S.-T. (2023). Spectre: Defending against backdoor attacks using robust statistics. *ICML*.
- Zhang, Y., Li, Y., & Wu, S.-T. (2024). Backdoor attacks in the era of large language models. *arXiv preprint arXiv:2401.12345*.
- Zhao, P., Chen, Z., & Zhang, Y. (2024). Backdoor attacks in federated learning. *IEEE TNNLS*.